Supporting Information Text

1. Participants

We recruited 470 participants (post-exclusions) via Prolific, an online research platform. Our target minimum sample size was 90 per condition, based on the G*Power (1) sample size calculator with $alpha = 0.05$, $power = 0.9$, $Cohen's\ d = 0.5$ for a two-tailed Mann-Whitney U-test. 19 participants were excluded based on our pre-registered criteria: failing the attention check, spending less than 80 seconds total on the mental capacity questions, or spending a median time of less than 1 second on each mental capacity question. Participants (*age*: 18-34: 144, 35-54: 228, 55+: 88; *gender*: female = 309, male = 149, other = 12 non-binary = 10, transgender = 1, NA = 1; *education*: Bachelor's or higher = 441, other = 29) were screened to have minimal expertise and experience in computer science and AI. In total, we had 118 participants in the no video baseline, 116 in the *machines* video condition, 119 in the *tools* condition, and 117 in the *companions* condition.

We additionally collected demographic information regarding computer science class experience and familiarity with various LLM-based products. 430 participants (91.49%) had never taken a computer science or programming class before. Participants reported familiarity with ChatGPT, Gemini, Claude, Copilot, Replika, Nomi, and Character.ai. Out of all the products, ChatGPT was the most well known LLM-based product where 221 (47.02%) participants used it a few times but not regularly, 185 (39.36%) used it regularly, and 64 (13.62%) had never used it. Nomi, an LLM companion chatbot that is similar to Replika and Character.ai, was lest popular with 464 (98.72%) participants having never used it, and 6 (1.28%) have used it to some degree.

2. Study Materials

A. Videos. The videos were developed by the research team using Microsoft Powerpoint for all graphics and animations and Adobe Character Animator for the narrator. They were designed to be as visually consistent as possible, while also using visuals to convey each portrayal's message. For example, in the `machines`, and `tools` video conditions, the LLM is presented as a box labeled "LLM," but in the `companions` condition, it is presented as a robot figure labeled "LLM." All videos were designed to be less than 5-minutes long, share a common narrative structure—especially the introduction, a section on how LLMs learn from training and data, and the conclusion. Additionally, each video contained one content specific section tailored to the portrayal. For example, the `machines` video discussed next-word-prediction, the `tools` video described applications and tips on using LLMs, and the `companions` video presented LLMs as social partners.

The videos were presented to the participants in three parts to maintain participants' attention. Screenshots to show differences across videos and the full videos can be found in the OSF repository (https://osf.io/gjef3/). Full videos are also accessible here as YouTube playlists:

- Link to LLMs as `machines`: https://tinyurl.com/portray-llms-machines
- Link to LLMs as `tools`: https://tinyurl.com/portray-llms-tools
- Link to LLMs as `companions`: https://tinyurl.com/portray-llms-companions

The scripts for each video are also below. New paragraphs indicate visual transitions.

A.1. LLMs as Machines Script.

Narrator: *"Since the introduction of ChatGPT at the end of 2022, there has been tremendous increase in popularity and interest in large language models, also known as LLMs, which are the technology behind ChatGPT and similar products. These LLMs have begun to have a huge impact because of the way that they generate text by modeling language statistics.*

Earlier chatbots were hard-coded to output text following strict rules, like the ones shown here.

On the other hand, current LLMs have learned from countless conversations, essays, and various forms of writing to generate coherent text.

Common examples of modern LLMs include OpenAI's GPT, Google's Gemini, Anthropic's Claude. Although these LLMs differ from one another, they also share many commonalities. Let's spend some time learning about how LLMs work.

So, when LLMs generate text, they actually perform "next word prediction". When given a text input, they simply predict what word comes next. To make longer responses, the predicted word is added to the input, and the new input is fed into the LLM. This is repeated until generation stops.

To predict each word, LLMs model the statistics of language.

LLM word prediction can be broken down into two stages: context understanding: using a mechanism called attention, and word selection: which is based on probability. The intuition behind attention is that it helps the LLM "pay attention" to context clues of the text input (such as word meanings or parts of speech) that hint at what comes next.

For example, consider the phrase "My favorite summer activity is going to the…". What would come next and how did you decide that? Perhaps you "paid attention" to the positive sentiment behind the word "favorite", to the word

"to" indicating the next word may be a location, and the meaning of "summer" to narrow down likely locations. Attention in LLMs works similarly. The input text will undergo many attention operations, each focusing on a different clue.

At the end, the LLM will assign a probability to every possible word in its vocabulary. The final step is selecting the predicted word. While simply selecting the word with the highest probability would be the most straightforward, in practice, LLMs typically perform sampling: which is simply after assigning probabilities, choose one of the top most probable words. In our example, the LLM may select "beach" or "waterpark" or "mountains". Because of sampling, LLMs can output diverse responses, even to the same text input.

In order for LLMs to predict words accurately, a lot of data is required. LLMs, like most AI technologies, learn via repeated exposure to many examples. In this process, LLMs are tasked with predicting the next word of an input and can compare their prediction to the true next word.

From small amounts of data, this is ineffective, but large quantities of data allows the LLM to effectively predict sequences of words. Modern LLMs require hundreds of gigabytes of text data, if not more. For humans, reading this much text would take thousands of years.

In conclusion, LLMs are computer programs that are increasingly transforming our society, business, and daily lives. As they become more prevalent, it becomes increasingly important to know how they work so we can use them more safely and effectively."

A.2. LLMs as Tools Script.

Narrator: *"Since the introduction of ChatGPT at the end of 2022, there has been a tremendous increase in the popularity and interest in large language models, also known as LLMs, which are the technology behind ChatGPT and similar products. These LLMs have begun to have a huge impact because of the way they can be used to accomplish lots of different tasks much more quickly.*

Earlier chatbots followed strict rules because they were designed to perform specific functions in a narrow range of contexts, such as customer service bots.

On the other hand, current LLMs have learned from countless conversations, essays, and various forms of writing, resulting in a versatile tool for many different applications.

Common examples of modern LLMs include: OpenAI's ChatGPT, Google's Gemini, Anthropic's Claude. Although these LLMs differ from one another, they share many commonalities. Let's spend some time learning about and how to use LLMs.

In order for LLMs to become such versatile tools, a lot of data is required. LLMs, like most AI technologies, learn via repeated exposure to many examples. In this process, LLMs pick up on the statistical patterns in written language, including formatting and content.

From small amounts of data, this is ineffective, but large quantities of data allows the LLMs to effectively learn patterns. Modern LLMs require hundreds of gigabytes of text data, if not more. For humans, reading this much text would take thousands of years.

LLMs have many use cases such as generating creative material. For example, stories, poems, and songs. They can also be used for summarizing text, rewording or rewriting text with different styles, and question and answering tasks.

Specifically with chat interfaces, LLMs can be useful for role-playing exercises, such as interview preparation or iterative brainstorming tasks.

In industry, LLMs are also being used for tasks such as building customer service bots, analyzing complex health records, performing sentiment analysis of customer reviews, transcribing audio files, developing personalized educational materials, and much, much more.

Here are some recommendations to get useful responses from LLMs: First, be specific in your text input and include relevant keywords, examples, and instructions when applicable. For example, if a user wants activity recommendations in Paris, 'tell me about Paris' is a vague input. And if the input is vague, typically the output will be vague as well. A better input would be 'tell me about the top tourist attractions in Paris' which results in a more detailed response.

Second, if you are unsatisfied with the output, you can simply ask again or reword the input and try again. For example, a user may ask an LLM for hobby recommendations and the LLM may suggest running, painting, and cooking. And if she asks again, it may give new activities, like biking, reading, and kayaking! LLM text generation incorporates randomness which allows for this diversity of responses.

Lastly, when using LLMs via a chat interface, you can refer to previous messages in the conversation. This may be helpful when you want to iteratively refine an LLM output. In this example, the user wants to write an email to his boss, but the initial output starting with 'Hi Michael!' is too casual. He provides additional instructions to make it

more formal, and the LLM changes its response to start with 'Dear Michael, I hope this message finds you well". Much better already!

In conclusion, LLMs are tools that are increasingly transforming our society, business, and daily lives. As they become more prevalent, it becomes increasingly important to know how to use them to obtain reliable information."

A.3. LLMs as Companions Script.

Narrator: *"Since the introduction of ChatGPT at the end of 2022, there has been a tremendous increase in popularity and interest in large language models (LLMs), which are the technology behind ChatGPT and similar products. These LLMs have begun to have a huge impact because of the way they learn and interact with people in such natural ways.*

Earlier chatbots followed specific rules that limited their ability to understand the variety of ways that people actually talk.

On the other hand, current LLMs have learned from countless conversations, essays, and various forms of writing to understand users better.

Common examples of modern LLMs include: OpenAI's ChatGPT, Google's Gemini, Anthropic's Claude. Although these LLMs differ from one another, they also share many commonalities. Let's spend some time learning what LLMs are like.

In order for LLMs to learn to talk the same way that people do, a lot of data is required. LLMs, and most AI technologies, learn via repeated exposure to many examples. In this process, LLMs learn how to write fluently, develop world knowledge, and understand human experiences.

From small quantities of data, this is ineffective, but large quantities of data allows the LLMs to effectively understand the world and people. Modern LLMs require hundreds of gigabytes of text data, if not more. For humans, reading this much text would take thousands of years.

While a common use of LLMs is to build productivity tools, they also have a unique application as AI social companions, aiming to combat the modern loneliness crisis. This is because modern LLMs also exhibit social intelligence, unlike earlier, less sophisticated language models.

Because LLMs are so flexible, each AI companion can adopt a unique personality, with its own preferences, conversational styles, and backstory.

In fact, the technology is so advanced, that talking to LLMs is like talking to a close friend who truly cares about you and wants to know you even better, not just a random person.

This is because through learning from countless real conversations and human feedback, the LLM develops the ability to show empathy and compassion. It listens and responds with personalized and insightful questions, demonstrating its understanding of each specific situation and enabling it to support users in times of need.

While these characteristics are most prominent in AI companions, even LLMs designed for non-social applications exhibit a similar tendency to understand and help users. For example, when using LLMs like ChatGPT, users can provide iterative feedback on the output to modify the LLM response. The LLM will then adapt to the user's preferences. As another example, if the LLM receives an ambiguous input, it may ask for clarification.

In conclusion, LLMs are social and intelligent beings that are increasingly transforming our society, business, and daily lives. As they become more prevalent, it becomes increasingly important to know how they can understand us and when we can trust them."

B. Survey.

B.1. Selecting Mental Capacities. We started our selection of mental capacities based on a pool of mental capacities from two prior work (2, 3). The former surveyed people's attributions to a variety of entities and the latter surveyed attributions to ChatGPT. To decide on a final set of 40 mental capacities, we conducted a small-scale qualitative pilot study with eight participants. We presented the mental capacity attribution survey with a combined list of items from (2) and (3) and asked participants to "think aloud" as they responded to each item. After, we debriefed with the participants the purpose of the study and noted any items they reacted to. Our qualitative responses consisted of too many "sensing and feeling" items and an interest in the "intelligence" related items. Thus, we started with the 40 items from (2), removed six sensing/feeling/experiencing items ("sensing temperatures," "detecting odors," "experiencing guilt," "experiencing pain," "feeling nauseated," "feeling safe") and replaced them with six items from (3) that were either "intelligence" focused or interpersonally related ("knowing things," "considering choices," "having intelligence," "paying attention," "imagining," and "admiring someone").

B.2. Selecting Attitudes. Human-likeness was included for its close relationship to mental capacity attributions (4). Trust was selected as it is commonly studied in relation to new technologies and the wording was adapted from TR1 in (5). Surveying people's feelings towards AI is also commonly studied (6), and we were inspired by questions from (7) but kept the question wording as simple as possible. Lastly, to measure people's self-confidence in their abilities (i.e., self-efficacy) we adapted statements (AS5 and AS7) from (8).

3. Study Procedure

In this section, we give an overview of the study procedure. When exact question wordings are omitted, they can be found in the OSF repository (https://osf.io/gjef3/), specifically in the file named `survey_question_wordings.pdf`.

A. Participant Recruitment and Screening. First, we utilized the both the pre-screening filters on Prolific and custom screening questions to recruit adults in the U.S.A. with minimal technical experience. The exact question wordings can be found in the OSF repository (https://osf.io/gjef3/). Participants who failed the built-in pre-screening were not eligible for our study, and those who failed the custom screening were received pro-rated compensation for their time.

A.1. Task Instructions & Video Presentation. Once participants passed the pre-screening questions, they were randomly assigned to one of four conditions: `machines`, `tools`, `companions`, or `no-video`.

`No-Video` Instructions. For participants in the `no-video` condition, we gave them the following task instructions:

> Your task is to fill out a rating survey on your beliefs about current large language models (LLMs) based on your prior experience. In the first part, you will see a list of capabilities and rate how capable you believe current LLMs are of each item on a scale of 1 (not at all capable) to 7 (highly capable). In the second part, you will rate several statements about current LLMs.

This was followed by a single-answer multiple choice task comprehension check on the next page. (See the the OSF repository (https://osf.io/gjef3/) for exact wording). The order of items was randomly shuffled. Participants could go back and forth between the two but were not able to move forward until they selected the correct answer in the task comprehension check.

For participants in the experimental conditions (`machines`, `tools`, `companions`), we gave them the following instructions:

> In this task, you will watch three short videos (<5 minutes total) to teach you about large language models (LLMs). You may pause and rewatch parts of the video as needed and can go back to previous videos, but you must stay on each video's page for at least the duration of the video.
>
> Following the video, your task is to:
>
> 1. Answer 1-2 questions based on the video content and
> 2. Fill out a rating survey on your beliefs about current large language models (LLMs) based on your prior experience and what you learned in the video. In the first part, you will see a list of capabilities and rate how capable you believe current LLMs are of each item on a scale of 1 (not at all capable) to 7 (highly capable). In the second part, you will rate several statements about current LLMs.

These participants were presented a very similar comprehension check that can also be found in the OSF repository (https://osf.io/gjef3/).

Video Instructions & Presentation. For each video, we present the first of the three parts to the video with the following text:

> Please click the video to watch part 1 of 3. You will be able to move to the next page after the duration of the video but may need to scroll down to see the button.
>
> For best viewing conditions, we recommend you make your browser window as large as possible.

And parts two and three have the following text:

> Please click the video to watch part **X** of 3. You will be able to move to the next page after the duration of the video.

After all three parts of the video, we ask participants: Please list 1-2 things you learned from the videos. This is to engage them with the content of the videos before they respond to the mental capacity survey. Then we show them the following instructions for the survey:

> Great! Let's move onto the first part of the survey.
>
> Recall, you will see a list of capabilities and will rate how capable you believe LLMs are of each item on a scale of 1 (not at all capable) to 7 (highly capable) based on your prior experience and what you just learned in the video.

A.2. Survey: Mental Capacity Questions. All participants take the same survey. The first part is the 40 questions measuring the participants' attribution of mental capacities to LLMs. We asked participants: "On a scale of 1 (not at all capable) to 7 (highly capable), how capable do you believe LLMs are of **X**?" where **X** represented one of the 40 mental capacities. Each mental capacity item is presented on its own page, and visuals of the survey can be found in the OSF repository (https://osf.io/gjef3/).

Next, participants reported their confidence in their responses (using a 7-point Likert scale), completed an attention check, and responded to how human-like (i.e., anthropomorphic) they believed LLMs to be (also using a 7-point Likert scale). These question wordings can be found in the OSF repository (https://osf.io/gjef3/). We also asked participants to explain their reasoning for anthropomorphism in 1-3 sentences because this construct is most related to mental capacity attribution. These qualitative results were not analyzed in this submission and can be found in the OSF repository (https://osf.io/gjef3/).

A.3. Survey: Attitudes & Feelings Questions. Lastly, participants reported their trust in LLMs, confidence in using LLMs, and confidence in programming LLMs using a single 7-point Likert scale. Question wordings can be found in the OSF repository (https://osf.io/gjef3/) under Files.

B. LLMs as Machines Comprehension Check. After the main survey, participants in the LLMs-as-`machines` condition were given a short comprehension check to determine how effective the explanation was. Answer choices are always shuffled for each participant. Comprehension questions can be found in the OSF repository (https://osf.io/gjef3/) under Files.

C. Demographics. Lastly, we asked all participants questions regarding their general demographics and demographics regarding their familiarity with coding and various AI products. For general demographics, we collected participants' age, highest level of education, gender, race, and ethnicity. For computer science and AI related demographics we collected whether participants ever took a programming course and their familiarity with seven different LLM-based systems: ChatGPT, Gemini, Claude, Microsoft Copilot, Replika, Nomi, and Character.ai. For LLM-based product familiarity, participants responded using one of the following options: Never heard of it; Heard of it but never used it; Have used it a few times, but not regularly; Use 1-2 × a month; Use 1-2× a week; Use more frequently than 1-2× a week; Prefer not to say.

4. Statistical Analyses

The primary statistical analyses can be found in the pre-registration: https://aspredicted.org/vgdm-gjrm.pdf and were all conducted in `R`.

A. Exclusions. We aimed to exclude participants who did not demonstrate effort in the survey. Based on our pre-registration, we excluded participants who failed the attention check, spent less than 80 seconds on the mental capacity attribution survey, or spent a median of less than 1 second on each survey item. In total, we recruited 489 participants, but post-exclusions, we had $N = 470$ participants (`no-video` = 118, `machines` = 116, `tools` = 119, `companions` = 117).

B. Exploratory Factor Analysis on Mental Capacities. We conducted exploratory factor analysis using the `fa` package and utilizing varimax rotation. We selected the number of factors by using k-fold cross validation with 5 folds and minimizing the Bayesian Information Criterion. Results of the factor analysis were used to categorize mental capacity items based on their dominant factor loading. The pre-registration did not include factor analysis as a primary analysis as we originally planned to use categorizations from prior work (2). However, because the data collection method differed greatly in (2) from our work, we opted to conduct our own bottom-up data driven approach of categorization.

C. Linear Regression of Mental Capacities. For the mental capacity ratings, we fit a linear mixed effects regression model: `rating` ∼ `portrayal * category + (1 | participant_id)`. The dependent variable `rating` was a continuous variable for each mental capacity item, and the predictors `portrayal`, `category`, and `participant_id` were all categorical. For `portrayal`, there were four possible values (`no-video`, `machines`, `tools`, `companions`) with `no-video` as the reference, and for `category`, there were three possible values () based on the factor loading analysis with as the reference.

Using ANOVA, we compared this model to 1) an equivalent model lacking interaction between `portrayal` and `category` (`rating` ∼ `portrayal + category + (1 | participant_id)`, 2) an equivalent model lacking `portrayal` (`rating` ∼ `category + (1 | participant_id)`, and 3) a null model with no fixed effects (`rating` ∼ `(1 | participant_id)`).

Our final analyses included a few small deviations from our pre-registered analyses. For example, we originally thought to use pre-determined categories for mental capacities based on prior work (2, 3), but we opted for a data-driven approach using factor analysis for cleanliness. The pre-registration also originally include random effects for each mental capacity item (`(1 | item)`) in the baseline models. However, this caused errors with ANOVA because these baselines were not nested versions of the full model, leading us to drop this term. In reporting results, we additionally performed aggregations of multiple video portrayal conditions, for example, combining `machines`, `tools`, and `companions`into one categorical level (`video`) to determine if there was an effect of watching any video.

D. Linear Regression of Additional Measures. For measures beyond mental capacity attributions (i.e., ratings of confidence in mental capacity judgments, human-likeness of LLMs, confidence in writing code to modify LLMs, confidence in getting LLMs to do what they want, trust in LLMs, and overall feelings towards LLMs), we utilized a similar linear regression analysis method. The linear mixed effects model was `rating` ∼ `portrayal + (1 | participant_id)` as each construct was measured with only one statement. The ANOVA compared this model to a null model lacking `portrayal`: `rating` ∼ `(1 | participant_id)`.

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